

Phase 0

data/NonFatal-OnePager.csv

**gitignored

Raw Data straight from RedCap Db



split_data.py

src/split_data.py

Patient's Profile Info

versioned/specimen_v1.csv

**gitignored

- 3 have Trailing spaces, 4 have real problems
- 22 Clean Columns, 7 have problems
- 373 rows * 29 cols

> specimen_v1.csv
- Just separated from the combined dataset

- Age : not uniform formatting, infants logged as 1/2 years. 370 plain integers, 3 are non-standard strings. Whole column has years, for infants needs months. Record of 7 (138) and 1 (157) month
- BAC (Blood Alcohol Cone) : Has 61 distinct values; Use of Two unit systems : g/dL, mg/dL, Ambiguous bare integers with no units, 56 rows have inequalities (<5, <0.010), 13 rows with text mixed with values (Ethanol <0.010), plus formatting issues with integers and Floats
- Sample Matrix : 100% empty, filled version in qtof_v1.csv
- Hospital Stay : Mostly clean integers, 25 rows have until (days) listed, some have inequalities (<1, >2), fractions (0.4), a full date 2022-08-09 (rec_3).

Patient's Lab Report

versioned/qtof_v1.csv

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> qtof_v1.csv
- Just separated from the combined dataset

- 396 rows, 5 cols, 373 unique patients
- 23 patients have 2 screens
- Positive Ion Mode : Free-text detections
- Negative Ion Mode : Free-text detections
- Free-Text Detections problems :
 - a. Trailing-comma
 - b. Missing Delimiters : (Analyte names joined by spaces)

data/Analyte_Category_Mapping.csv

Analyte_Category_Mapping

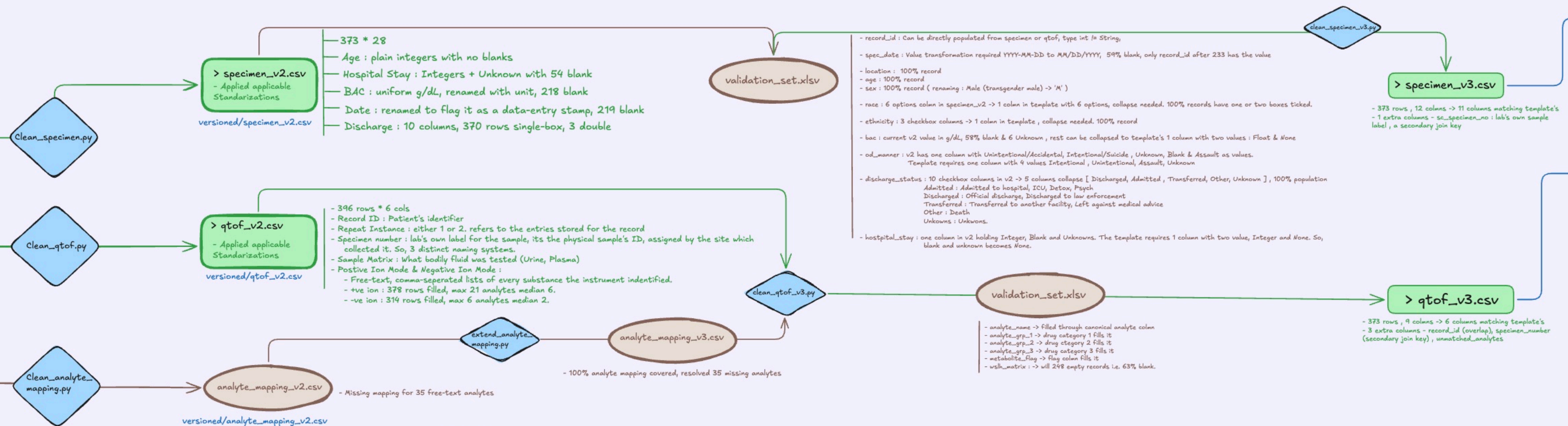
- Category Typos :
 - a. DC1 : 16 distinct values, 4 misspellings
 - b. DC2 : NSPOpioids - NPSOpioids
 - c. DC3 : Hallucinogen - Hallucinogens
- Duplicate Row : Drug appears twice, one has Flag = Metabolite and other has Flag = Blank.
- Casing and Trailing inconsistency

STANDARDIZATIONS

- Age :
 - any under 12 months -> 0
 - full integer Column
 - BAC :
 - target unit = g/dL
 - if result > 1, divide by 1000 and record that with g/dL
 - if result < 1, record the value in g/dL
 - if reported with below inequality, record the value as 0.
 - Hospital Stay
 - ignore any full date values
 - for values between 0-1, record as 1 day
 - if values can't be reasonably converted, record it as Unknown
 - whole column should be integer value of days.
-
- * ANALYTE MAPPING CORRECTIONS
- NarcoticAnalgesics
 - NarcoticAnalgesics
 - CNSStimulants
 - DissociativeAnesthetics
- Missing :
duplicate drug clarification for Flag
-
- data/validate.csv

Template format for Specimen record modification provided

Phase 1



- 444 rows * 7 cols

- Analyte : 444 spellings/possible ways someone might type the 297 drugs.

- Canonical_Analyte : Actual spellings of 297 drugs.

- Known_variants : Nicknames or Abbreviations people use for drugs.

- Drug_Category_1 : Broad family the drug belongs to.

- Drug_Category_2 : More Specific family the drug belongs to.

- Drug_Category_3 : More Specific family the drug belongs to.

- Flag : Is this the drug, or its left overs

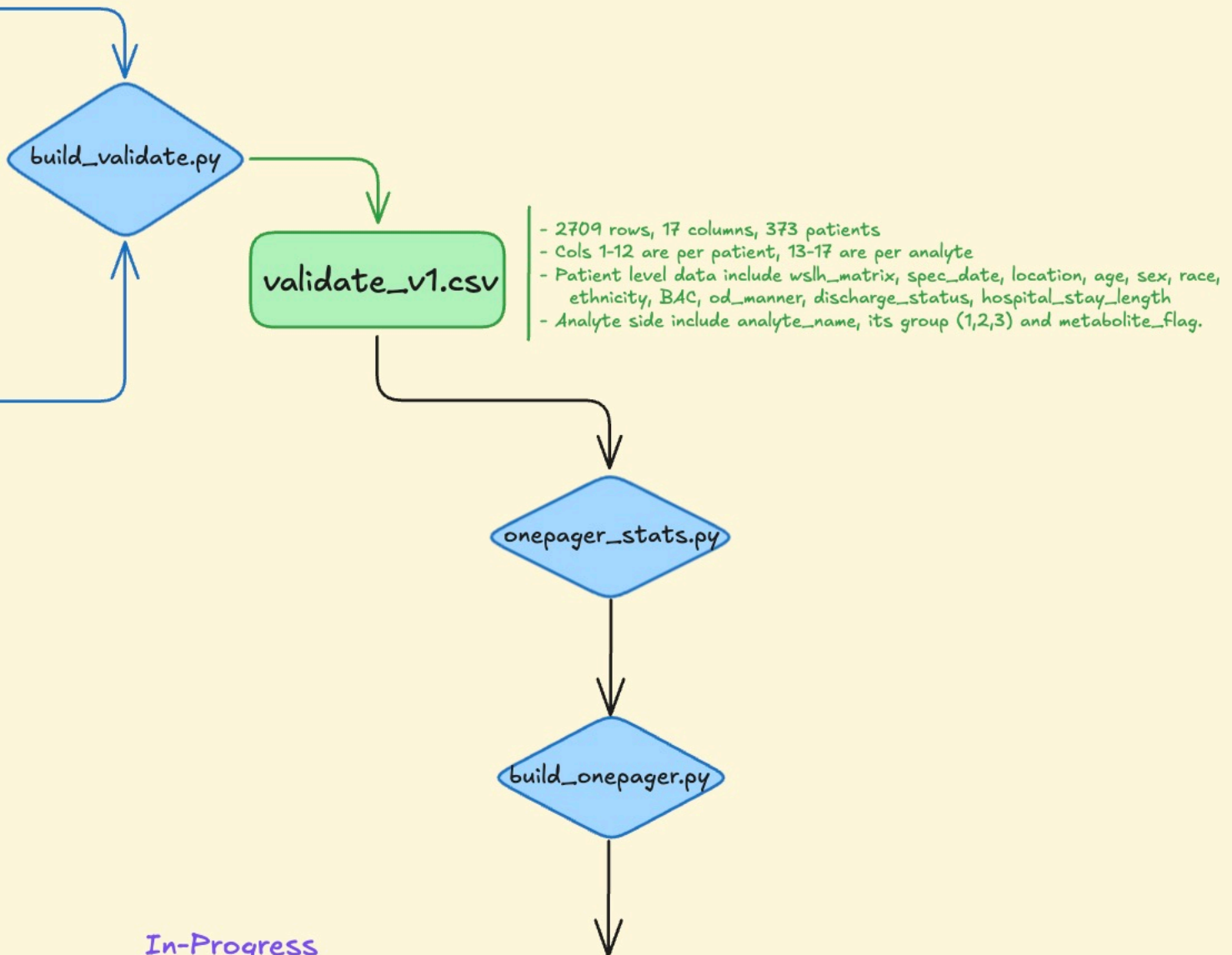
- When your body processes a drug, it breaks it into byproducts, and the instrument detects those too. Flag=Metabolite means "this row is a breakdown product, not the drug itself."

Drug_Cat_1		Drug_Cat_2		Drug_Cat_3	
rows	distinct drugs	rows	distinct drugs	rows	distinct drugs
- NarcoticAnalgesics :	63	42	- NPSOpioids :	34	23
- Antidepressants :	48	28	- Cathinone :	16	14
- CNSStimulants :	47	35	- Amphetamines :	11	7
- Benzodiazepines :	31	21	- Cocaine :	11	7
- Cannabinoids :	30	7	- Fentanyl :	7	3
- Antipsychotics :	28	18			
- DissociativeAnesthetics :	12	6			
- MOUD :	10	7			
- MuscleRelaxers :	6	5			
- Barbiturates :	3	3			
- Hallucinogens :	2	2			
- Xanaxone :	2	2			

- Hallucinogens : 1 rows, 1 distinct drugs -> CNSStimulants -> Amphetamines

- record_id : Can be directly populated from specimen or qtof, type int != String,
- spec_date : Value transformation required YYYY-MM-DD to MM/DD/YYYY, 54% blank, only record_id after 233 has the value
- location : 100% record
- age : 100% record
- sex : 100% record (renaming : Male (transgender male) -> 'M')
- race : 6 options coln in specimen_v2 -> 1 coln in template with 6 options, collapse needed. 100% records have one or two boxes ticked.
- ethnicity : 3 checkbox columns -> 1 coln in template, collapse needed. 100% record
- bac : current v2 value in g/dL, 58% blank & 6 Unknown, rest can be collapsed to template's 1 column with two values : Float & None
- od_manner : v2 has one column with Unintentional/Accidental, Intentional/Suicide, Unknown, Blank & Assault as values. Template requires one column with 4 values Intentional, Unintentional, Assault, Unknown
- discharge_status : 10 checkbox columns in v2 -> 5 columns collapse [Discharged, Admitted, Transferred, Other, Unknown], 100% population
Admitted : Admitted to hospital, ICU, Detox, Psych
Discharged : Official discharge, Discharged to law enforcement
Transferred : Transferred to another facility, Left against medical advice
Other : Death
Unknowns : Unknowns.
- hospital_stay : one column in v2 holding Integer, Blank and Unknowns. The template requires 1 column with two value, Integer and None. So, blank and unknown becomes None.

Phase 2



In-Progress

INFOGRAPHICS GENERATION